Harsha Venkateshwara

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EDUCATION

University at Buffalo **Aug 2025 – Dec 2026**
Master's, Computer Science and Engineering

Visvesvaraya Technological University **Jun 2018 – Jul 2022**
Bachelor's, Mechanical Engineering

SKILLS

- **Programming & Data**: Python, C++, NumPy, Pandas, Scikit-Learn, OpenCV, Linux, Git
- **Frameworks**: PyTorch, JAX/Flax, TensorFlow, Ray, DeepSpeed, FSDP, Hugging Face, vLLM, TensorRT, Orbax, Grain
- **GPU & Systems**: CUDA C/C++, Triton, NCCL, MPI, XLA, Nsight Systems/Compute, Kernel Fusion, Memory Optimization
- **Cloud & Infra**: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Slurm, Ray Cluster, TensorBoard
- **Engineering & Development**: Object-Oriented Design, Data Structures & Algorithms, Distributed Systems, Data Engineering (ETL), MLOps, Cloud-Native Architecture, Agile Methodologies

EXPERIENCE

Itron, Inc. **Aug 2023 – Aug 2025**
Technical Associate – Data Systems & Analytics

- Architected production ML pipelines for energy and utility analytics, supporting **8–10M+ smart-meter and sensor records** across operational and analytical workflows.
- Engineered regression, classification, and time-series models for load forecasting and anomaly detection, achieving **18–30% performance gains** over statistical baselines.
- Streamlined large-scale data processing through advanced feature engineering, reducing data noise by **25%** and improving downstream model stability in production.
- Automated data ingestion, training, evaluation, and retraining workflows in Python, cutting **manual analysis effort by 35%** and accelerating experimentation cycles by 2×.
- Collaborated with product managers, domain experts, and engineers to translate **ambiguous operational requirements into scalable ML solutions** supporting data-driven decisions.
- Implemented continuous model validation and monitoring strategies, maintaining **greater than 90% accuracy** under evolving data distributions.

Kyono Software Pvt. Ltd. **Jun 2022 – Jul 2023**
Machine Learning Engineer

- Delivered end-to-end ML solutions across **6–8 client projects** in e-commerce, healthcare, and retail, owning model development from data preparation to deployment.
- Implemented recommendation, classification systems that improved personalization and operational insights by **10–20%**.
- Built scalable preprocessing and feature engineering pipelines handling **100K–1M+ records** from heterogeneous data sources.
- Integrated and optimized models in production systems with backend and frontend teams, reducing inference latency by **14%** without sacrificing accuracy.

Bharat Electronics Limited **Sep 2021 – Oct 2021**
Data Science Intern – Analytics and Reporting: Air Force Defense Unit

- Supported defense analytics initiatives for Indian Air Force systems in a **mission-critical, security-restricted environment**, contributing to operational data analysis and reporting.
- Analyzed and visualized **sensitive operational datasets**, adhering to **100% security and documentation compliance** while following high-reliability practices for accuracy, validation, and traceability.

PROJECTS

Electric Smart Grid Stabilizer – Itron, Inc. hackathon https://github.com/harsha-venkateshwara/smart-grid-stability-project

- Built a machine learning pipeline to process large-scale telemetry data, performing data normalization, feature engineering, and label construction for model training.
- Evaluated scikit-learn baselines and deep learning models in PyTorch, analyzing performance metrics and sensitivity to input features to support model fine-tuning and real-world deployment considerations.

Identification of Cancer-Specific Genetic Signatures https://github.com/harsha-venkateshwara/icmr-Cancer-gene-expression

- Processed high-dimensional datasets using Python and statistical methods, applying dimensionality reduction and supervised classification techniques.
- Compared classical machine learning models and deep neural networks to assess performance and generalization behavior.